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Macroeconomic conditions and bank failure

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Abstract

Utilizing a simple time-varying hazard model, we incorporate nationwide and state-level economic variables with banking-industry and bank-level data to examine U.S. bank failures during 1977–2019. We find that bank-level financial conditions are more essential in predicting bank failure, although macro factors affect the failure likelihood of vulnerable banks. We also find that banking-industry market variables are significant predictors. Unlike bank systemic funding cost whose predictive power is subsumed in the presence of macroeconomic variables, banking-industry market performance has a significantly independent predictive power on bank failure. This finding is novel to existing literature of bank-failure forecast and has important policy implications.

KEYWORDS

bank failure, failure prediction, hazard model, macroeconomic conditions

1 | INTRODUCTION

One major objective of regulators and policymakers is to ensure the stability and soundness of banking system, given the essential role of banking sector in a country's economic activities. Bank failure generally is contagious (Aharony & Swary, 1996; Pino & Sharma, 2019; Rajan & Ramcharan, 2016). In a mild case, it would spark a shock to the supply of credit, which would result in financial constraints and reduce employment at related companies (Benmelech et al., 2019; Farinha et al., 2019; Ghosh, 2017). In severe circumstances, bank failure contagion may induce a financial crisis that would subsequently exert an independent negative real effect (Campello et al., 2010; Dell'Ariccia et al., 2008; Kupiec & Ramirez, 2013) and cause severe economic disruption[1] (Boyd et al. 2005; Hoggarth et al., 2002; Hutchison & Noy 2005; Serwa 2010). Thus, the development of early warning models of bank failures is of great interest to regulators seeking to identify vulnerable banks and take prompt corrective actions to avoid bank failures.

In this paper, we estimate a time-varying hazard model of U.S. bank failures over the period 1977–2019, where we augment a standard set of bank-level explanatory variables with both nationwide and state-level economic variables as well as banking-industry market variables. Our sample period encompasses two waves of bank failures—one in the late 1980s and the other one in the late 2000s—as well as periods of economic growth and periods of economic recession, which enables us to test the impact of market and economic variables on bank-failure forecasts.

The prediction of bank failures has been extensively examined in literature. Early empirical research utilizes logit, probit, or static hazard models that treat the health of a bank as a function of its financial condition taken from a single point of time (e.g., Cole & Gunther, 1995; Cole & White, 2012; Lane et al., 1986; Martin, 1977;

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Whalen, 1991). Later empirical studies employ time-varying models that utilize explanatory variables observed at multiple points of time. One of the most popular models is the Cox time-varying hazard model (e.g., Brown & Dinc, 2005; Brown & Dinc, 2011; Mannasoo & Mayes, 2009; Molina, 2002; Whalen, 1991; Wheelock & Wilson, 2000). Shumway (2001) proposes a simple discrete time-varying hazard model that can be estimated using logistic techniques. The Shumway model has been widely used to predict corporate bankruptcy (Bharath & Shumway, 2008; Bonfim, 2009; Campbell et al., 2008; Chava & Jarrow, 2004; Nam et al., 2008), but has been applied to bank failure prediction only in more recent empirical studies (Betz et al., 2014; Cole & Wu, 2009; Cole & Wu, 2017; Hong & Wu, 2013). Audrino et al. (2019) introduce a mixed-data sampling (MIDAS) weighting scheme into the logit model to predict U.S. bank failures and find that the MIDAS logit model improves the predictive accuracy for longer-term forecasting from six to eight quarters ahead, although its estimation process is complicated.

In this study, we utilize Shumway (2001)'s hazard model for the following reasons. First, unlike the Cox time-varying model that uses partial-likelihood approach to estimate from samples of failed banks and comparable non-failed banks, Shumway (2001)'s model enables us to analyze the entire population of U.S. banks over the sample period by utilizing the full-information maximum likelihood approach, which is much more tractable and more efficient than partial-likelihood estimation (see Efron, 1977). Second, as demonstrated in a proof by Shumway (2001), this model can be estimated using standard logit techniques, although it does require the adjustments of test statistics to account for the lack of independence between firm-year observations. Third, by using time-varying covariates, we can incorporate market and macroeconomic variables to investigate the impact of market and macroeconomic factors on individual bank failures.

Existing related empirical studies mostly rely upon bank-level accounting data, occasionally augmented with market-price data (e.g., Balla et al., 2019; Cole & Gunther, 1995; Cole & White, 2012; Martin, 1977; Meyer & Pifer, 1970; Pettway & Sinkey, 1980), to develop early warning indicators of bank failure. These indicators are closely related to supervisory rating systems for banks. The most well-known rating system is the CAMEL system, which stands for *C*apital Adequacy, *A*sset Quality, *M*anagement, *E*arnings, and *L*iquidity.[2] The empirical research on U.S. bank failures primarily analyzes the bank-level data from waves of bank failures that happened during the late 1980s (Cole & Gunther, 1995, 1998; Wheelock & Wilson, 2000) or late 2000s (Cleary & Hebb, 2016; Cole & White, 2012; Cox & Wang, 2014; DeYoung & Torna, 2013). Cole and Wu (2017) find that a parsimonious specification performs well in predicting bank failures over the 1985–1993 period and during 2009–2010, respectively. They suggest that the characteristics of "distressed banks" have experienced little change despite substantial changes of the U.S. banking industry in terms of technology, regulations, and business activities over the past two decades. Balla et al. (2019) examine bank failures and failure costs in both failure periods and find that failure rates were lower, but failure costs were higher in the more recent crisis.

Distinguished from the existing research that aims to develop early warning models using bank-level data, our study focuses on how macroeconomic and market factors affect bank failure. Our research contributes to the existing literature in the following aspects.

First, to our best knowledge, our sample period is the longest one among the existing empirical studies, which enables us to investigate the impact of market and macroeconomic factors on bank failure and add new insight on bank-failure forecast that cannot be achieved by previous empirical studies using short sample periods. Our research is significantly different from Balla et al. (2019) who also incorporate macroeconomic variables into their study on bank failure. First, they examine failures among community banks over two discrete short sample periods, 1986–1992 and 2007–2013; we investigate bank failures among the whole bank population in the U.S. over a continuous long sample period 1977–2019. Second, they use state macroeconomic variables while we incorporate both national and state macroeconomic variables as well as banking-industry market variables. Third, they employ a static probit model while we utilize a time-varying hazard model. Thus, our findings complement theirs and have broader implications. Existing studies either examine a short sample period covering one wave of bank failures or analyze two short sample periods covering both waves of bank failures separately (e.g., Balla et al., 2019; Cole & Wu, 2017). Our sample period covers two waves of bank failures as well as economic booms and recessions in the U.S.; and for the first time, we use the coefficients estimated from the in-sample period covering the first wave of bank failures to investigate the predictive accuracy of the out-of-sample period covering the second wave of bank failures. We split our long continuous sample period into an in-sample period of 1977–2005 and an out-of-sample period of 2006–2019 so that each sample period includes one wave of bank failures and spans both good and bad economic times. We also examine two discrete short sample periods, an in-sample period of 1985–1992 and an out-of-sample period of 2009–2011, so that each period includes one wave of bank failures, respectively. We find that the coefficients of traditional proxies for CAMEL estimated from the in-sample period 1977–2005

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(1985–1992) correctly identify 92.2% (89.2%) of failed banks among the top 5% worst predicted probabilities in the out-of-sample period 2006–2019 (2009–2011). The results provide powerful new empirical evidence in support of the supervisory/regulatory policy focusing on banks' traditional risk ratios, particularly those related to capital adequacy, asset quality, liquidity, and profitability. Our results also highlight that using longer sample periods covering both crisis and non-crisis times improves the accuracy of forecasting bank failures than using short sample periods covering banking crisis times only.

Second, we investigate the impact of market and macroeconomic conditions on bank failures over time, during crisis and non-crisis times, respectively. We find that bank-level financial variables are more essential in predicting bank failure while banking-industry market variables also have significant predictive power. Macroeconomic conditions, particularly state-level economic conditions, can significantly affect overall bank failures. However, our results show that models using either banking-industry market variables alone or macroeconomic variables alone do not predict bank failures accurately, although the coefficients of some variables are statistically significant. Furthermore, the predictive power of both state and national economic variables dissipates in the presence of bank-level financial variables, while banking-industry market variables retain significant predictive power. We confirm the finding of Hong and Wu (2013) that bank systemic funding cost is a significant predictor of bank failure, but provide new empirical evidence that its predictive power depends on macroeconomic conditions as its significant predictive power vanishes once we include macroeconomic variables into the regressions. Unlike systemic funding cost, *bank excess return*, our proxy for banking-industry market performance, has a significantly independent predictive power on bank failure regardless of macroeconomic and market conditions. This finding is novel to existing bank failure literature and has important policy implications since information on bank stock returns is publicly available.

Third, we investigate how macroeconomic shocks affect bank failure through bank-level financial conditions. We find that national economic recession will exacerbate asset-quality problems and significantly increase the likelihood of failure for those banks with high percentage of non-performing loans. This finding holds given general market and macroeconomic conditions but does not hold conditioned on economic recession as economic recession itself has already significantly increased the probability of bank failure. We also find that, on one hand, a shock of interest rates makes those banks heavily relying on purchased funds more susceptible to failure; on the other hand, it also reduces the failure likelihood of those banks holding large percentage of liquid assets. However, this effect of interest-rate shock is not significant given macroeconomic and bank-level financial conditions. Although our empirical evidence on how macroeconomic shocks affect bank failure via bank-level financial conditions is inconclusive yet, our research is a starting point and opens up fruitful avenues for future research.

The remainder of our manuscript is organized as follows. Section 2 describes our data and measurement of variables. Section 3 briefly explains the econometric models. Section 4 presents our empirical results, while Section 5 provides a summary and conclusion.

2 | DATA AND MEASUREMENT OF VARIABLES

2.1 | Data

We obtain bank financial data from the year-end Reports of Condition and Income, informally known as the "Call Reports." Call Reports are filed by all FDIC-insured commercial banks with the Federal Financial Institutions Examination Council ("FFIEC"), which collects bank information on behalf of the three primary U.S. banking regulators—the Federal Deposit Insurance Corporation ("FDIC"), the Federal Reserve System ("FRS"), and the Office of the Comptroller of the Currency ("OCC").[3] The data include detailed balance-sheet and income-statement information of individual banks over the period of 1976–2018. We acquire information on the identities and closure dates of individual banks over the period of 1977–2019 from the FDIC website,[4] as well as the U.S. bank price index and national macroeconomic information from Datastream International. The data of state-level GDP growth rates and state-level unemployment rates are downloaded from the websites of the Bureau of Economic Analysis[5] and the U.S. Bureau of Labor Statistics,[6] respectively. Table 1 presents the detailed data sources and the variable definitions/measurements.

Following existing empirical studies, we use year-end bank financial variables to predict bank failures that occurred during the subsequent year. We match bank financial data reported at the end of year t with the information of failed banks in year $t + 1$. We use year-end bank financial data, instead of quarterly data, so that we can construct consistent and comparable bank financial variables and compare our results with existing research. Most of the failed banks have the previous year-end Call Reports available even if they failed during the first quarter of the following year. We drop 18 failed banks without

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TABLE 1 Variable definitions and data sources.

Variables	Definitions/measurements	Data sources
Capital ratio	Total equity capital divided by total assets: RCFD3210/RCFD2170	Bank call reports
Earnings ratio	Net income divided by total assets: RIAD4340/RCFD2170	Bank call reports
Non-performing loan ratio (NPL)	Sum of past loan due and non-accrual loans as a ratio of total assets: (RCFD1407 + RCFD1403)/RCFD2170	Bank call reports
Other real estate	Other real estate owned as a ratio of total assets: RCFD2150/RCFD2170	Bank call reports
Liquid asset ratio	The sum of cash and total investment securities (TIS) divided by total assets: (RCFD0010 + TIS)/RCFD2170. TIS = RCFD0400 + RCFD0600 + RCFD0900 + RCFD0380-RCFD1000 for the period 1976–1983, it equals to RCFD0390 for the period 1984–1993, and the sum of RCFD1754 and RCFD1773 for the period 1994–2018.	Bank call reports
Large CDs to assets	Time deposits $100,000 or more as a ratio of total assets: RCON2604/RCFD2170. Derivation formula from 2010 the RCON2604 equals to the sum of RCONJ473 and RCONJ474.	Bank call reports
Expense ratio	Total operating expenses as a ratio of total assets: RIAD4130/RCFD2170. If RIAD4130 is not available (mostly in later few years in the sample period) then RIAD4130 = RIAD4073 + RIAD4093.	Bank call reports
Size	Natural log of total assets: Ln (RCFD2170)	Bank call reports
Bank excess return	The difference between the continuous bank index return and the three-month T-bill rate.	Datastream International
Systemic funding cost (sys_cost)	The spread between the average three-month interbank rate and the three-month T-bill rate.	Datastream International
Growth rate	The continuous real annual GDP growth rate in the U.S.	Datastream International
Inflation rate	The annual changes of consumer price index in the U.S.	Datastream International
Unemployment rate	The annual unemployment rate in the U.S.	Datastream International
State growth	The annual GDP growth rates in the states of the U.S.	Bureau of Economic Analysis
State unemployment	The average of monthly unemployment rates over the year in the states of the U.S.	U.S. Bureau of Labor Statistics
Dgrowth	The dummy variable that takes a value of one when the U.S. GDP growth rate is negative, otherwise takes a value of zero.	N/A

Note: This table presents the data sources and the detailed measurements of variables, including the calculation of bank financial variables using the MDRM number. For example, RCFD2170 stands for total assets in the call reports. As not all banks report every financial item in the call reports over such a long period, we use the data dictionary on the website of the Board of Governors of the Federal Reserve System to check the definitions of mnemonic codes in order to construct consistent financial variables over the sample period.

the previous year-end call reports from the sample. Our sample period covers 43 years, during which the Mnemonic (series) codes of bank financial items in the Call Reports have been adjusted several times. We then carefully check the definitions and availability of Mnemonic codes in the data dictionary on the website of the Board of Governors of the Federal Reserve System[7] to construct consistent variables over such a long period.[8] This long sample period includes two waves of bank failures as well as both economic boom and recessionary periods, which facilitates our investigation on how market and macroeconomic conditions affect bank failures.

2.2 | Bank financial variables

We follow extant literature to construct traditional proxies for CAMEL using bank-level financial data. Numerous previous studies have found that these variables are significant predictors of bank failures (e.g., Balla

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et al., 2019; Cole & Gunther, 1995, 1998; Cole & White, 2012; Demirguc-Kunt, 1989; DeYoung & Torna, 2013; Gajewski, 1989; Martin, 1977; Shaffer, 2012; Thomson, 1992; Whalen, 1991).

2.2.1 | Capital adequacy

Capital adequacy is the key variable considered important for ensuring the health of banks. We use a simple capital ratio, measured by the ratio of total equity capital to total assets, in this study to ensure the consistency of measurement given the length of our sample period.[9] Empirical studies have shown that the simple capital adequacy ratio is a significant predictor of the U.S. bank failures over the recent crisis (Balla et al., 2019; Cole & White, 2012; Cole & Wu, 2017; DeYoung & Torna, 2013; Shaffer, 2012). Both Mayes and Stremmel (2013) and Chernkykh and Cole (2015) find that, relative to risk-weighted capital ratios, a non-risk-weighted adequacy measure does a better job in predicting U.S. bank failures over 1992–2012. Bank capital can absorb shocks of unexpected losses and preserve the confidence of banks. Thus, we expect that capital adequacy is negatively associated with the probability of bank failure.

2.2.2 | Asset quality

Since the dominant business of commercial banks is lending, we measure bank asset quality focusing on non-current loans. We construct two variables measuring the proportion of nonperforming assets to total assets. The first one is *non-performing loan ratio*, measured by the sum of *past-due loans*[10] *and nonaccrual loans* divided by total assets. The second one is *other real estate*, measured by *other real estate owned*[11] as a ratio of total assets. These variables reflect poor bank asset quality, so we expect they have a positive association with the likelihood of bank failure.

2.2.3 | Management

The competency of management plays a crucial role in bank performance. However, it is difficult to measure the competency or quality of bank management. The FDIC provides an efficiency ratio that reflects expenses as a percentage of revenues to assess management quality (Mayes & Stremmel, 2013). Consistent with FDIC, we use the ratio of total operating expenses to total assets as an inverse proxy for management efficiency. We expect that the *expense ratio* is positively related to the likelihood of bank failure.

2.2.4 | Earnings

We use the *earnings ratio*, defined as net income divided by total assets, to measure bank profitability. The more profitable is a bank, the less likely it will fail. We expect the *earnings ratio* to have a negative relation with the likelihood of bank failure.

2.2.5 | Liquidity

Liquidity is essential for a bank to meet its short-term financial obligations, especially unexpected withdrawals of deposits. We measure bank liquidity using two different variables—one on the asset side and the other one on the liability side. The asset-side liquidity variable is *liquid assets ratio*, measured by the sum of cash and investment securities divided by total assets. Banks mostly hold government bonds as investment securities. Investment securities are far more liquid than loans, and they enable banks to minimize fire-sale losses in response to unexpected demands of cash. Therefore, we expect *liquid assets ratio* to be negatively related to the likelihood of failure. Our liability-side liquidity proxy is *large CDs (certificates of deposit) to assets*, which is measured by large certificates of deposit ($100,000 or more) divided by total assets. These types of deposits are highly volatile, in that they are the first to be withdrawn when a bank is rumored to have financial difficulties. In addition, banks that heavily rely upon purchased funds, such as large CDs, rather than core deposits, often have a more aggressive liquidity management strategy and face higher funding costs. Hence, we expect that *large CDs to assets* will be positively associated with the likelihood of failure.

Most of the banks in our sample are privately held; thus it is implausible to construct a variable consistently measuring banks' *sensitivity to market risk*. This sixth component of CAMELS has been widely neglected in existing empirical studies due to the difficulty to construct a proper proxy using bank accounting and financial data. Therefore, in addition to the above proxies for CAMEL, we construct a variable *bank size*[12] as a standard control variable. Small banks are generally more vulnerable to failure because their asset portfolios are less well diversified, so we expect that *bank size* will be negatively related to the probability of failure. Table 2 summarizes the expected sign of the coefficient for each variable.

Since troubled banks tend to have more non-standard financial variables, applying a standardized percentage to winsorize outliers may cut off those banks from the sample as well as the useful information from the estimations.[13] Thus in this paper, we report the results on the unwinsorized sample that includes all available

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observations. Our full sample contains 449,087 observations covering more than 22,000 banks, out of which 2,033 banks failed during 1977–2019.[14]

Figure 1 shows the percentage of failed banks to all banks for each year from 1977 to 2019. There were two waves of failures during the sample period. The first one occurred during the late 1980s and peaked in 1988 with 1.8% of banks failed, and the second one happened during the recent global financial crisis, peaking in 2010 with 1.9% of banks failed. Very few banks failed during 1995–2004, and no bank failed in the years 2005, 2006, and 2018.

TABLE 2 Expected sign of bank financial variables in predicting bank failure.

Variables	Measurements	Expected sign
Capital adequacy	Capital ratio	–
Asset quality	Non-performing loan ratio	+
	Other real estate	+
Management	Expense ratio	+
Earnings	Net earnings ratio	–
Liquidity	Liquid asset ratio	-
	Large CDs to assets	+
Bank size	Natural log of total assets	–

Note: This table summarizes the expected sign of CAMEL variables and bank size in the empirical models of predicting bank failures.

Figure 2 shows the banks that failed in each year as a percentage of the 2,015 banks that failed during the entire sample period. The early wave of failures peaked in 1988 at about 12.6% while the later one peaked in 2010 at around 6.7%.

Table 3 reports the summary statistics of bank financial variables and the correlation matrix of main variables. Panels A–C report the mean values and standard errors of bank financial variables for the full sample banks, failed banks, and non-failed banks over the full sample period 1976–2018 (Panel A) and two sub-sample periods (i.e., 1976–2004 in Panel B, and 2005–2018 in Panel C), respectively. The differences in the mean values of non-failed and failed banks are presented in the last column of each panel. As expected, failed banks generally have lower capital adequacy, profitability, asset quality, liquidity, and management efficiency. Each of the mean differences is statistically significant at the 1% level. Compared with non-failed banks, failed banks are generally smaller, have lower capital ratio, earnings ratio and liquid asset ratio, and higher non-performing loan ratio, other real estate owned ratio, large CDs to assets ratio and expense ratio. However, as indicated in Panels B and C, failed banks in the first wave of bank failures were smaller banks, while it was the opposite in the second wave of bank failures. We conjecture that this may be resulted from the dramatic consolidation of the banking industry.[15]

Panel D of Table Error! Reference source not found reports the Pearson's correlation matrix of the primary variables in our regressions. "Fail" is the dummy variable that takes a value of one for a bank classified by FDIC as

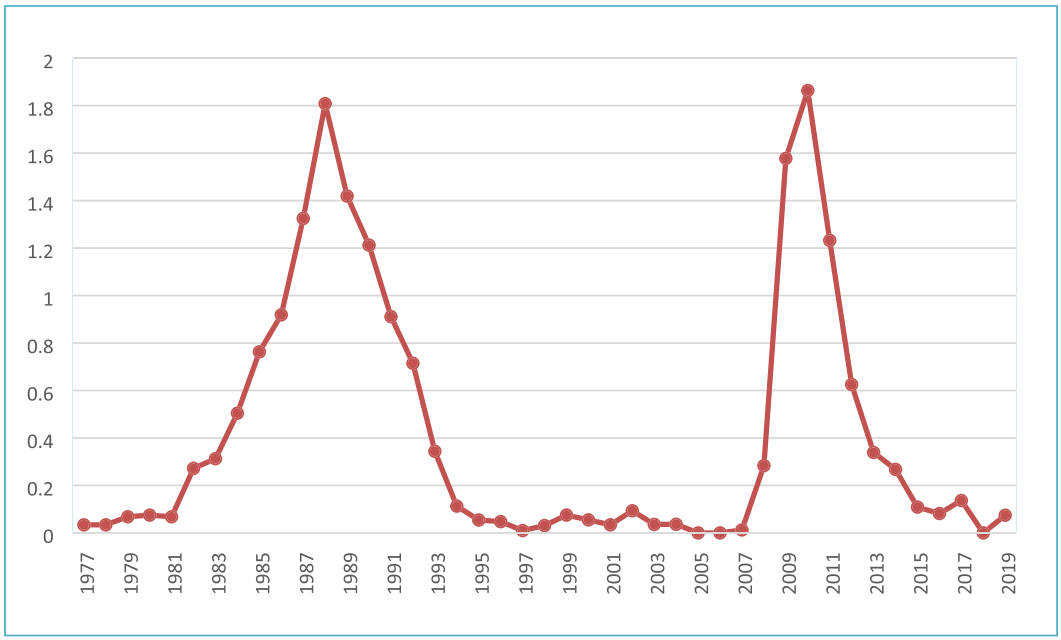

FIGURE 1 Failure percentage of banks. This figure presents, for each year, the proportion of failed banks among our sample banks.

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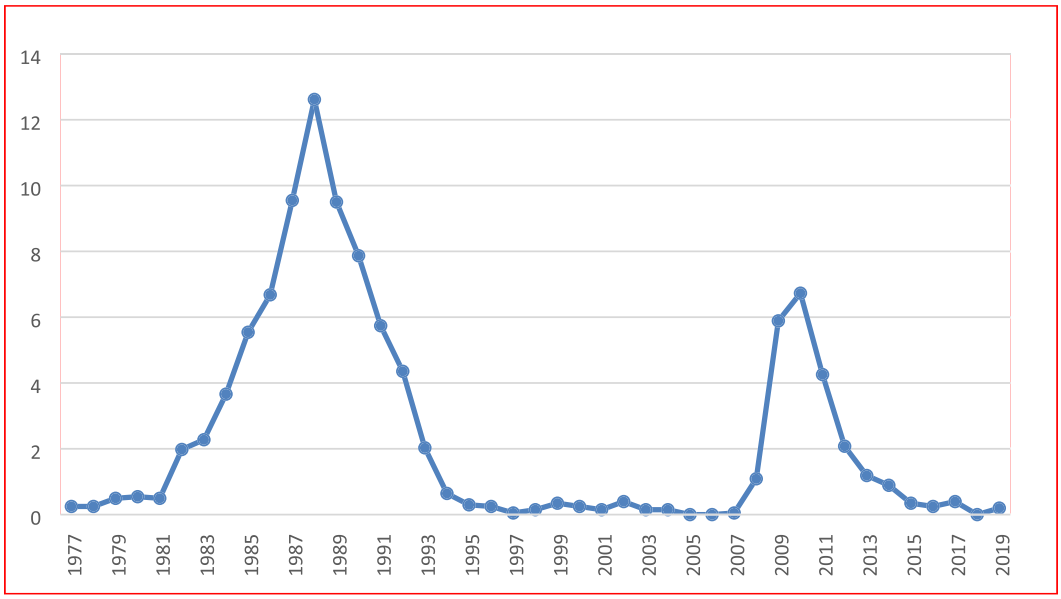

FIGURE 2 Percentage of total failures. This figure reports, for each year, the percentage of failed banks among the total number of 2,015 banks that failed during 1977–2019.

a failed bank, otherwise zero. All variables are significantly correlated with bank failure. Notably, several bank-level financial variables are significantly correlated with national and state economic variables. The unreported correlation matrixes for the 1977–2005 and 2006–2019 sub-sample periods show similar results, except that the correlation coefficient between *bank size* and *Fail* is significantly negative for the first subsample period but turning to the opposite for the second subsample period, which is consistent with the results reported in Panels B and C.

2.3 | Market and macroeconomic variables

We incorporate market and macroeconomic variables into the hazard model and examine the impact of market and macroeconomic factors on bank failures over time. We use economic variables to proxy for general macroeconomic conditions at nationwide level, including *U.S. GDP growth rate, U.S. inflation rate*, and *U.S. unemployment rate*, and at the state level, including *State GDP growth rate* and *State unemployment rate*, respectively. We also construct two market-based macro variables specific for banking industry.

Our first banking macro variable is *bank excess return*, measured by subtracting the risk-free rate from the annual bank index return. Previous research has documented that past stock returns are strongly related to bankruptcy (Campbell et al., 2008; Curry et al., 2007; Pettway, 1980; Pettway & Sinkey, 1980; Shumway, 2001). Listed banks can be a broad representative of a country's banking sector, and bank portfolio excess returns are significantly linked with a country's future economic growth (Cole et al., 2008). As most of our sample banks are privately-held banks for which market data do not exist, we then use *bank excess return* as a proxy for the market performance of banking sector.

Our second banking macro variable is *bank systemic funding cost (sys_cost)*. Systemic risk has become a topical issue among both researchers and bank regulators since the occurrence of the recent global financial crisis. Bisias et al. (2012) provide a review of 31 quantitative measures of systemic risk in the literature of finance and economics. According to the taxonomy of Bisias et al. (2012), bank funding risk and shock transmission can be regarded as an ex-ante systemic risk measure and be used as an early-warning indicator of crisis. Interbank spreads have been utilized as a proxy for bank systemic liquidity risk in recent empirical research (e.g., Cornett et al., 2011; Hong & Wu, 2013). Hong and Wu (2013) find that bank systemic funding risk, measured by the spread between the three-month London offered interbank rate (LIBOR) and the three-month U.S. Treasury bill rate, is a major predictor of U.S. bank failures during 2008–2009. Since LIBOR is only available from 1986, and also due to the recent controversy about the integrity of LIBOR (Abrantes-Metz et al., 2012; Fouquau & Spieser, 2015), we use the spread between the average three-month interbank rate and the three-month T-bill rate as a proxy for bank systemic funding risk. We also construct an

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TABLE 3 Summary statistics and correlation matrix.

Panel A: Summary statistics of bank financial variables 1976–2018

	Failed banks			Non-failed banks			Full sample			Mean diff.
Variables	**NOB**	**Mean**	**Std error**	**NOB**	**Mean**	**Std error**	**NOB**	**Mean**	**Std error**	**Nonfailed-failed**
Capital ratio	25,077	0.0859	0.0004	424,083	0.1021	0.0001	449,160	0.1012	0.0001	0.0161***
Earnings ratio	25,077	−0.0013	0.0002	424,083	0.0089	0.0000	449,160	0.0083	0.0000	0.0102***
Non-performing loan ratio	25,077	0.0164	0.0002	424,083	0.0068	0.0000	449,160	0.0073	0.0000	−0.0096***
Other real estate	25,077	0.0087	0.0001	424,082	0.0033	0.0000	449,159	0.0036	0.0000	−0.0054***
Liquid assets ratio	25,077	0.2838	0.0009	424,064	0.3450	0.0002	449,141	0.3415	0.0002	0.0612***
Large CDs to assets	24,622	0.1605	0.0008	420,954	0.1064	0.0001	445,576	0.1094	0.0001	−0.0541***
Expense ratio	24,927	0.0844	0.0002	422,845	0.0704	0.0002	447,772	0.0712	0.0001	−0.0140***
Size	25,077	10.7602	0.0099	424,083	11.0961	0.0022	449,160	11.0773	0.0022	0.3359***

Panel B: Summary statistics of bank financial variables 1976–2004

	Failed banks			Non-failed banks			Full sample			Mean diff.
Variables	**NOB**	**Mean**	**Std error**	**NOB**	**Mean**	**Std error**	**NOB**	**Mean**	**Std error**	**Nonfailed-failed**
Capital ratio	22,661	0.0854	0.0005	330,667	0.0960	0.0001	353,328	0.0953	0.0001	0.0106***
Earnings ratio	22,661	−0.0001	0.0002	330,667	0.0090	0.0000	353,328	0.0085	0.0000	0.0091***
Non-performing loan ratio	22,661	0.0141	0.0002	330,667	0.0065	0.0000	353,328	0.0070	0.0000	−0.0076***
Other real estate	22,661	0.0079	0.0001	330,666	0.0030	0.0000	353,327	0.0033	0.0000	−0.0049***
Liquid assets ratio	22,661	0.2961	0.0009	330,666	0.3629	0.0003	353,327	0.3586	0.0002	0.0668***
Large CDs to assets	22,206	0.1532	0.0008	327,538	0.0944	0.0001	349,744	0.0981	0.0001	−0.0588***
Expense ratio	22,511	0.0870	0.0002	329,429	0.0758	0.0002	351,940	0.0765	0.0002	−0.0111***
Size	22,661	10.5900	0.0099	330,667	10.8198	0.0024	353,328	10.8051	0.0023	0.2298***

Panel C: Summary statistics of bank financial variables 2005–2018

	Failed banks			Non-failed banks			Full sample			Mean diff.
Variables	**NOB**	**Mean**	**Std error**	**NOB**	**Mean**	**Std error**	**NOB**	**Mean**	**Std error**	**Nonfailed-failed**
Capital ratio	2,416	0.0909	0.0014	93,416	0.1236	0.0003	95,832	0.1228	0.0003	0.0327***
Earnings ratio	2,416	−0.0123	0.0007	93,416	0.0085	0.0001	95,832	0.0080	0.0001	0.0208***
Non-performing loan ratio	2,416	0.0382	0.0010	93,416	0.0078	0.0000	95,832	0.0086	0.0001	−0.0304***
Other real estate	2,416	0.0164	0.0006	93,416	0.0044	0.0000	95,832	0.0047	0.0000	−0.120***
Liquid assets ratio	2,416	0.1682	0.0022	93,398	0.2815	0.0006	95,814	0.2787	0.0006	0.1134***
Large CDs to assets	2,416	0.2276	0.0025	93,416	0.1485	0.0003	95,832	0.1505	0.0003	−0.0791***
Expense ratio	2,416	0.0608	0.0006	93,416	0.0513	0.0004	95,832	0.0515	0.0004	−0.0095***
Size	2,416	12.3568	0.0279	93,416	12.0740	0.0046	95,832	12.0811	0.0044	−0.2828***

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Panel D: Correlation matrix of main variables.

	Fail	Capital ratio	Earnings ratio	NPLs ratio	Other real estate	Liquid assets ratio	Large CDs to assets	Expense ratio	Size	Bank excess return	Systemic funding cost	Growth rate	Inflation rate	Unemployment rate	State growth	State unemployment
Fail	1															
Capital ratio	−0.076***	1														
Earnings ratio	−0.210***	0.206***	1													
NPLs ratio	0.316***	−0.141***	−0.295***	1												
Other real estate	0.236***	"-0.010***	−0.226***	0.416***	1											
Liquid assets ratio	−0.055***	0.093***	0.117***	−0.148***	−0.123***	1										
Large CDs to assets	0.067***	0.052***	−0.126***	0.099***	0.104***	−0.243***	1									
Expense ratio	0.040***	0.216***	0.256***	0.050***	0.031***	0.027***	−0.034***	1								
Size	−0.004***	−0.126***	0.028***	0.031***	0.015***	−0.236***	0.186***	−0.107***	1							
Bank excess return	−0.019***	0.001	0.029***	−0.057**	0.010***	0.038***	−0.041***	0.001	−0.023***	1						
Systemic funding cost	0.015***	−0.044***	−0.024***	0.125***	0.011***	0.061***	0.007***	0.117***	−0.113***	−0.051***	1					
Growth rate	−0.011***	−0.051***	0.017***	−0.021***	−0.049***	0.069***	−0.111***	−0.002	−0.136***	0.097***	−0.311***	1				
Inflation rate	−0.013***	−0.116***	0.028***	−0.091***	−0.076***	0.125***	−0.132***	0.057***	−0.321***	−0.237***	0.011***	0.017***	1			
Unemployment rate	0.020***	−0.082***	−0.051***	0.199***	0.092***	0.163***	−0.031***	0.062***	−0.164***	0.123***	0.360***	−0.23***	0.103***	1		
State growth	−0.026***	−0.090***	0.038***	−0.050***	−0.080***	0.134***	−0.148***	0.015***	−0.240***	0.024***	−0.213***	0.202***	0.437***	0.196***	1	
State unemployment	0.047***	−0.067***	−0.092***	0.230***	0.152***	0.127***	0.049***	0.062***	−0.021***	0.076***	0.281***	−0.158***	0.035***	0.670***	0.033***	1

Note: Panels A–C of this table present the numbers of observations (NOB), the means and standard errors of bank financial variables for the failed banks, non-failed banks, and all sample banks over the full sample period (1976–2018) and two subsample periods (1976–2004 and 2005–2018), respectively. The differences in means between the variables of non-failed banks and failed banks are presented in the last column. The folded F statistics rejected the null hypothesis that the variable variances for two groups are equal, so we use the Satterthwaite *t*-statistics assuming unequal variance of variables. Results based upon pooled *t*-statistics assuming equal variance are qualitatively the same.

Panel D presents the Pearson's correlation matrix of the main variables used in the regressions. "Fail" is the dummy variable that takes a value of one for a bank classified by FDIC as a failed bank, otherwise zero. ***, **, and * indicate different significance levels to reject the null hypothesis that the correlation coefficient is zero.

***Indicates the significance level at 1%;

**Indicates the significance level at 5%;

*Indicates the significance level at 10%.

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alternative variable using the spread between the three-month LIBOR and three-month T-bill rate to check for robustness. The results are not reported but are qualitatively similar.

3 | ECONOMETRIC MODEL

In this section, we briefly explain how we utilize the simple time-varying hazard model to analyze bank failures; a detailed description and proof of this model can be found in Shumway (2001). This time-varying hazard model treats a firm's health as a function of its latest financial condition and can be estimated using logit techniques with appropriate adjustment of test statistics. More importantly, this model enables us to incorporate time-varying market and macroeconomic variables into the analyses and examine the impact of macroeconomic factors on bank failures over a long sample period.

We assume that a bank can only fail at a discrete point of time, $T_i = 1, 2, 3, \ldots$ We define a dummy variable Y_i that is equal to one if a bank failed during year T_i, and otherwise zero. Let $F(t_i, X; \gamma)$ be the probability mass function of failure, where γ represents a vector of parameters and X represents a vector of explanatory variables. Following the hazard model conventions, the survivor function that gives the probability of surviving up to time T can be defined as:

$$S(T,X;\gamma) = 1 - \sum_{J<T} F(J,X;\gamma) \tag{1}$$

The hazard function giving the probability of failure at time T conditional on surviving to time T can be expressed as:

$$H(T,X;\gamma) = \frac{F(T,X;\gamma)}{S(T,X;\gamma)} \tag{2}$$

The likelihood function of the discrete-time hazard model is given by:

$$L = \prod_{i=1}^{n} H(T_i,X_i;\gamma)^{Y_i} S(T_i,X_i;\gamma) \tag{3}$$

The discrete-time hazard model is equivalent to a multiple-period logit model with the following likelihood function (Cox & Oakes, 1984; Shumway, 2001):

$$L = \prod_{i=1}^{n} \left\{ H(T_i,X_i;\gamma)^{Y_i} \prod_{J<T_i} [1 - F(J,X_i;\gamma)] \right\} \tag{4}$$

where F is the cumulative density function (CDF) of failure that depends on time T and can be interpreted as a hazard function.

Therefore, this discrete-time hazard model can be estimated using a panel logit model with appropriate adjustment of the test statistics. The test statistics estimated from a panel logit model assume that the bank-year observations are independent. However, for the discrete-time hazard model, the bank-year observations of a particular bank cannot be independent because a bank cannot fail in period T if it failed in period T-1; similarly, a bank surviving to period T cannot have failed in period T-1. Thus, each bank's life span only makes one observation for the hazard model, so the correct number of independent observations is the number of banks rather than the number of bank-year observations. As a result, the correct test statistics of the hazard model can be derived by dividing the chi-square test statistics of a logit model by the average number of bank-year observations in the sample.

4 | EMPIRICAL RESULTS

4.1 | In-sample estimation and out-of-sample predictive accuracy

In this section, we perform the in-sample estimation and examine the out-of-sample forecast accuracy in order to confirm the validity of the traditional proxies for CAMEL over a continuous long sample period. Recent empirical studies indicate that, despite the substantial changes of the U.S. banking industry over the last two decades in terms of technology, regulations, and business activities, the characteristics of "distressed bank" have experienced little change. For example, Cole and White (2012) find that the traditional proxies for CAMEL components do an excellent job in explaining the U.S. commercial bank failures in year 2009. Shaffer (2012) finds that a standard set of CAMEL proxies is significant in both the early and late wave of failures but notes that the magnitudes of several coefficients were significantly different. Fahlenbrach et al. (2012) find that banks that performed poorly during 1998 when Russia defaulted some of its debt are prone to perform poorly during the recent financial crisis, providing empirical evidence in support of the persistence of a bank's risk culture. Cole and Wu (2017) find that a parsimonious model fitting the data from 1985 to 1993 using traditional proxies for CAMEL also performs well in predicting bank failures during 2009–2010. However, these existing empirical studies focus on the analyses of short sample periods during which many banks failed, and they do not use the coefficients estimated from the first

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wave of bank failures to predict failures over the second wave. Different from existing empirical studies, our sample period covers both waves of bank failures as well as times when few banks failed, so it is worthwhile to confirm the validity of our model specification in predicting bank failures over such a long sample period.

In the first step, we split our continuous sample period into an in-sample period of 1977–2005 and an out-of-sample period of 2006–2019 so that each sample period includes a major banking crisis, that is, one in late 1980s for the in-sample period and the other one in late 2000s for the out-of-sample period, respectively. Importantly, both also include years during which there were very few failures. The coefficients estimated from the in-sample period are combined with the year-end bank financial data over 2005–2018 to predict bank failures over the out-of-sample period of 2006–2019. We then sort all banks from the out-of-sample period based on their predicted failure probabilities calculated using the fitted coefficients from the in-sample estimation, and rank the predicted failure probabilities from highest/worst to lowest/best and calculate at each observation the percentage of failures that are correctly classified (1 – Type 1 Error Rate). We then report this percentage of correctly classified failures for various percentiles of the sorted banks.[16] The in-sample estimated results and the out-of-sample predictive accuracy are presented in Panel A of Table 4.

The in-sample estimation results in Panel A show that each of the coefficients of bank financial variables has the expected sign. Coefficients of the proxies for bank capital

TABLE 4 In-sample estimation results and out-of-sample predictive accuracy.

	Panel A			Panel B		
In-sample estimation results	**In-sample period 1977–2005**			**In-sample period 1985–1992**		
Variable	**Coefficient**	**Adjusted χ^2**	***P*-value**	**Coefficient**	**Adjusted χ^2**	***P*-value**
Intercept	1.69	1.31	0.252	1.35	1.52	0.218
Capital adequacy	−53.23	85.41	0.000	−54.21	164.02	0.000
Earnings ratio	−6.67	1.47	0.225	−1.73	0.19	0.660
Non-performing loan ratio (NPL)	18.45	25.61	0.000	18.07	43.37	0.000
Other real estate	13.57	8.50	0.004	10.08	9.99	0.002
Liquid assets ratio	−3.70	8.57	0.003	−3.11	12.14	0.000
Large CDs to assets	3.25	8.56	0.003	2.38	8.83	0.003
Expense ratio	2.98	0.97	0.324	3.86	1.58	0.209
Size	−0.35	9.00	0.003	−0.29	12.36	0.000
No. of failed banks	1,539			1,249		
No. of observations	349,695			108,024		
Out-of-sample predictive accuracy	**Out-of-sample period 2006–2019**			**Out-of-sample period 2009–2011**		
Worst predicted probability (%)	**Accuracy ratio**	**Cumulative accuracy**		**Accuracy ratio**	**Cumulative accuracy**	
0–5	92.2%	92.2%		89.2%	89.2%	
5–10	2.7%	95.0%		3.8%	93.0%	
10–20	1.9%	96.8%		4.1%	97.1%	
20–30	1.1%	97.9%		0.6%	97.7%	
30–40	0.2%	98.1%		1.5%	99.1%	
40–50	0.6%	98.7%		0.3%	99.4%	
50–100	1.3%	100.0%		0.6%	100.0%	

Note: This table presents the in-sample estimation results and the out-of-sample predictive accuracy. Panel A presents the estimation results over the in-sample period 1977–2005 and the predictive accuracy over the out-of-sample period 2006–2019. Panel B reports the estimation results over the in-sample period 1985–1992 and the predictive accuracy over the out-of-sample period 2009–2011. The results are estimated by using 1-year lagged bank financial variables. The chi-square values for the hazard model have been adjusted for the average number of bank years per bank. The coefficients estimated from the in-sample period are applied to lagged bank financial data to forecast bank failures over the out-of-sample period. The accuracy ratio is measured as the number of predicted bank failures among the probability rankings against the actual number of bank failures.

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adequacy, asset quality, liquidity, and bank size are all statistically significant at the 1% level. Smaller banks, banks with higher ratio of non-performing loans and higher percentage of other real estate owned, and banks relying more on large CDs for funding are more likely to fail, while banks with higher capital adequacy and more liquid assets are less likely to fail. The coefficients of *earnings ratio* and *expense ratio* are not statistically significant, which indicates that the marginal impact of profitability or operating efficiency is negligible.[17] The out-of-sample predictive accuracy ratio is measured as the number of predicted failures among the probability rankings against the actual number of bank failures. Our model correctly identifies 92.2% (95.0%) of the failed banks among the worst 5% (10%) predicted failure probabilities. The results in Panel A confirm that the traditional CAMEL proxies do an excellent job in explaining the second wave of bank failures.

In the next step, we focus on the analyses of two short sample periods during which many banks failed.[18] A few existing studies also examine the out-of-sample predictive accuracy, but their in-sample and out-of-sample periods are all from the same wave of bank failures (Cole & White, 2012; Cole & Wu, 2017).[19] We examine the predictive accuracy of the out-of-sample period 2009–2011 based on the estimated results from the in-sample period 1985–1992. The number of failed banks during these two short sample periods makes up more than 80% of the total failures over our full sample period, with more than 4% of total failures in each year (see Figure 2). The results are presented in Panel B of Table 4. As shown in Panel B, the in-sample estimated results are qualitatively the same as their counterparts in Panel A; however, the out-of-sample predictive accuracy among the worst 5% (10%) predicted failure probabilities is 89.2% (93%), a reduction of 3% (2%) compared to its counterpart in Panel A. The result indicates that using longer sample periods could improve the predictive accuracy of bank failures.

Compared with existing empirical studies, our results provide more powerful evidence that bank early warning models based upon historical failures can serve regulators as important supervisory tools even at the very beginning of a new wave of bank failures. The results also provide convincing empirical evidence in support of the robustness of traditional CAMEL framework in predicting bank failures over a long sample period in the following sections.

4.2 | Macroeconomic conditions and bank failure

Recent empirical studies have shown that macroeconomic factors may significantly affect bank risk-taking (e.g., Buch et al., 2014; Calmès & Théoret, 2014; Khan et al., 2017). Distinguishing from the existing research, we examine the impact of market and macroeconomic factors on bank failures. Utilizing the data over the full sample period 1977–2019, we first estimate the hazard model using only our market, national, and state economic variables. The values of market and nationwide economic variables are the same for all banks in each year, that is, there is time-series variability but no cross-sectional variability. In contrast, the state economic variables vary across both years and states. We then estimate the hazard model using bank financial variables only, and then sequentially add the market, nationwide, and state variables to the model to examine the incremental impact of market and macroeconomic conditions on bank failures. We estimate a full sample including all banks and a subsample including relatively small banks whose size is below the median size of the full sample, and the results are presented in Tables 5 and 6, respectively.

Models 1–4 in Table 5 report the estimation results using market variables, nationwide economic variables, state economic variables, and all market and economic variables, respectively. As shown in model 1, the coefficient of *bank excess return* is negative while the coefficient of *systemic funding cost* is positive, and both coefficients are significant at the 5% level. The result indicates that poor overall market performance or high systemic funding costs of banking sector will signal a high probability of bank failures in the following year. The probability of bank failures is significantly lower under the macroeconomic conditions with higher national inflation and lower national unemployment rates, while the impact of national economic growth is negligible (model 2). As shown in model 3, banks in the states with lower economic growth and higher unemployment rates have a significantly higher chance of failing. However, when all market and economic variables are included in the regression (model 4), the coefficients of *bank excess return* and two state variables remain significant at the 1% level, while the coefficients of *systemic funding cost* and all national variables lose the significance. The result of model 4 suggests that *bank excess return* has an independent explanatory power, while the predictive power of *systemic funding cost* depends on economic conditions. Noticeably, the intercept coefficients of models 1–4 are all highly significant with a p-value close to zero, indicating the models may have missing essential variables. Although some individual explanatory variables are statistically significant, the model specifications are not sufficient to explain bank failures.

Model 5 in Table 5 presents the estimation results using bank financial variables only, while models 6–9 report the results of sequentially adding the variables of

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TABLE 5 Macroeconomic conditions and bank failure, 1977–2019.

Variables	1	2	3	4	5	6	7	8	9
Intercept	−5.52***	−6.51***	−6.62***	−6.48***	−0.19	−0.16	0.97	0.31	0.79
Capital ratio					−52.80***	−53.86***	−54.33***	−53.70***	−55.47***
Earnings ratio					−11.31**	−11.21**	−9.89**	−10.26**	−9.02*
Non-performing loan ratio (NPL)					19.14***	19.59***	18.77***	17.93***	18.31***
Other real estate					10.75***	11.54***	10.02**	10.15**	10.65**
Liquid asset ratio					−3.79**	−3.86**	−3.56***	−3.69***	−3.60***
Large CDs to assets					2.98***	2.89***	2.90***	2.72***	2.65**
Expense ratio					1.78	0.56	2.09	1.89	1.78
Size					−0.16	−0.17*	−0.21**	−0.20*	−0.23**
Bank excess return	−1.09**			−1.37***		−0.96*			−1.05*
Systemic funding cost (*sys_cost*)	12.17**			−2.15		22.56**			17.37
Growth rate		−2.98		3.48			−10.46		−3.48
Inflation rate		−8.79***		−5.56			−5.96		−6.55
Unemployment rate		17.61***		−6.66			0.35		0.78
State growth rate			−10.01***	−8.32***				−6.34**	−3.99
State unemployment rate			26.38***	31.71***				5.37	4.31
AUROC	0.551	0.606	0.73	0.739	0.976	0.977	0.974	0.974	0.975
No. of failed banks	2,033				2,015				
No. of observations	449,087				445,485				

Note: This table presents the estimation results using the time-varying hazard model. The results are estimated by using 1-year lagged bank financial variables, market and macroeconomic variables from 1976 to 2018 to predict bank failures over the period of 1977–2019. The chi-square values for the hazard model have been adjusted for the average number of bank years per bank. AUROC is the area under the ROC curve.
***Indicates a significance level at 1%;
**Indicates a significance level at 5%;
*Indicates a significance level at 10%.

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TABLE 6 Macroeconomic conditions and the failure of small banks, 1977–2019.

Variables	1	2	3	4	5	6	7	8	9
Intercept	−5.40***	−5.39***	−6.10***	−5.35***	4.42**	4.20**	3.99*	4.81**	4.31*
Capital ratio					−52.89***	−53.55***	−53.92***	−53.19***	−54.12***
Earnings ratio					−4.49	−4.22	−3.85	−3.99	−3.84
Non-performing loan ratio (NPL)					20.92***	21.44***	20.05***	20.10***	20.11***
Other real estate					14.25***	14.95***	14.59***	13.65***	14.72***
Liquid asset ratio					−3.56***	−3.62***	−3.64***	−3.54***	−3.60***
Large CDs to assets					3.62***	3.54***	3.46***	3.28***	3.28***
Expense ratio					1.97	1.61	1.91	1.94	1.67
Size					−0.64***	−0.63***	−0.64***	−0.68***	−0.65***
Bank excess return	0.055			−0.63		0.33			0.04
Systemic funding cost (*sys_cost*)	7.71			1.44		19.48**			17.58
Growth rate		7.28		10.30*			0.87		4.33
Inflation rate		−13.67***		−10.46**			−4.53		−3.74
Unemployment rate		6.59		−18.40*			11.61		3.87
State growth rate			−9.11***	−7.52***				−4.21	−2.94
State unemployment rate			20.51***	28.15***				6.16	2.3
AUROC	0.592	0.581	0.694	0.72	0.975	0.977	0.976	0.977	0.977
No. of failed banks	1,149				1,141				
No. of observations	236,982				232,787				

Note: This table presents the estimation results using the time-varying hazard model for a subsample of relatively small banks whose size is below the median size (size = 11) of the full sample. The results are estimated by using 1-year lagged bank financial variables, market and macroeconomic variables from 1976 to 2018 to predict bank failures over the period of 1977–2019. The chi-square values for the hazard model have been adjusted for the average number of bank years per bank. AUROC is the area under the ROC curve.

***Indicates a significance level at 1%;

**Indicates a significance level at 5%;

*Indicates a significance level at 10%.

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models 1–4 to the specification of model 5. As shown in model 5, each coefficient of the bank financial variables has the expected sign; except for the *expense ratio* and the *size* variables, all bank financial variables are statistically significant. When both banking market variables are added to the specification, *bank excess return* and *systemic funding cost* remain significant at the 10% and 5% levels, respectively (model 6). In contrast, when nationwide and state economic variables are included in the regression, respectively, as shown in models 7 and 8, only the coefficient of *state growth rate* retains a significance level of 5%, and all other national- and state-level economic variables lose the statistical significance. In model 9, the coefficient of *bank excess return* remains significant at the 10% level, while the coefficients of all other market and economic variables are not significant in the presence of bank financial variables, suggesting that the explanatory power of *bank excess return* is independent of bank financial variables and economic conditions. Models 6–9 show that the inclusion of market and economic variables improves the explanatory power of bank size variable whose coefficient becomes significant; bank financial variables proxied for capital adequacy, asset quality, earnings, and liquidity remain statistically significant regardless of the inclusion of market and economic variables. The results indicate that the effect of these bank financial variables on bank failure is independent of macroeconomic conditions, while the impact of bank size and *systemic funding cost* to some extent depends on macroeconomic conditions.

Figure 3 presents the receiver operating characteristic (ROC) curves of the models in Table 5. A ROC curve is a graphical illustration of the diagnostic capacity of a binary classifier system by plotting the true positive rate (TPR) against the false positive rate (FPR) at various threshold settings. The true-positive rate is known as sensitivity, while the false-positive rate is also known as the probability of false alarm and can be calculated as (1 − specificity). The predictive accuracy of a model specification is measured by the area under the ROC curve (AUROC). An area of 1.00 represents a perfect prediction; an area of 0.50 indicates no predictive power.

As a rough guide, we rank the predictive accuracy of a model as excellent, good, fair, poor, and fail based on the five deciles above 0.50. For example, a model will be ranked as "excellent" with the AUROC in between 0.90 and 1.00 and "fail" with the AUROC in between 0.50 and 0.60. Although the coefficients of both banking market variables are statistically significant at the 5% level, model 1, with an AUROC of 0.551, fails to predict bank failures accurately. The model using only nationwide macroeconomic variables also does a poor job in predicting failed banks, with an AUROC of only 0.606 for model 2. Instead, state economic variables do a fair job in predicting bank failures, with the AUROC of 0.730 for model 3. The predictive accuracy of model 4 is slightly improved by adding the banking market variables and nationwide variables to model 3, with an AUROC of 0.739 compared with 0.730 for model 3. In sharp contrast, model 5, which uses only bank-level financial variables, does an excellent job in predicting bank failures as indicated by an AUROC of 0.976. Its predictive accuracy does not improve by adding either national or state economic variables into the model, as shown by the AUROCs of models 7 and 8, respectively. Adding banking-industry market variables slightly enhances the predictive power of the model with an AUROC of 0.977 for model 6. In summary, Figure 3 confirms the findings of Table 5 and provides complementary findings about the predictive accuracy of the model specifications in Table 5.

We also investigate the impact of market and macroeconomic conditions on bank failures during crisis and non-crisis times, respectively.[20] We divide our full sample period into two discrete sub-periods: one for the crisis times covering two waves of bank failures (i.e., 1985–1992 and 2009–2011), and the other one for the non-crisis times (i.e., 1977–1984, 1993–2008, and 2012–2019). We then rerun the nine models in Table 5 for two sub-periods, respectively. The overall results[21] confirm that bank-level financial conditions are more essential in predicting bank failures in either crisis or non-crisis times. Except for *bank excess return* whose coefficient remains significant at the 5% level in crisis times, all other market and macroeconomic variables lose their explanatory power in the presence of bank financial variables. Except for model 1, the AUROC of each model in the sub-periods is lower than its counterpart in Table 5, indicating that using a sample period covering both crisis and non-crisis times may improve the predictive accuracy of bank failures.

In Table 6, we re-estimate the nine model specifications in Table 5 but include only small banks whose asset size is below the median of the full sample (i.e., natural log of assets < 11). Smaller banks constitute more than half of the bank failures during the full sample period. Distinguished from the counterparts in Table 5, both banking-industry market variables have no predictive power on the failures of small banks (models 1 and 4), which makes sense as small banks are mostly privately held. Models 2–4 show that the economic conditions of the state, where the smaller banks are headquartered and primarily operate, significantly have more impact on bank failure than the nationwide macroeconomic conditions. In model 5, the coefficients of all bank financial variables are statistically significant except for the *earnings ratio* and *expense ratio*. The results are consistent

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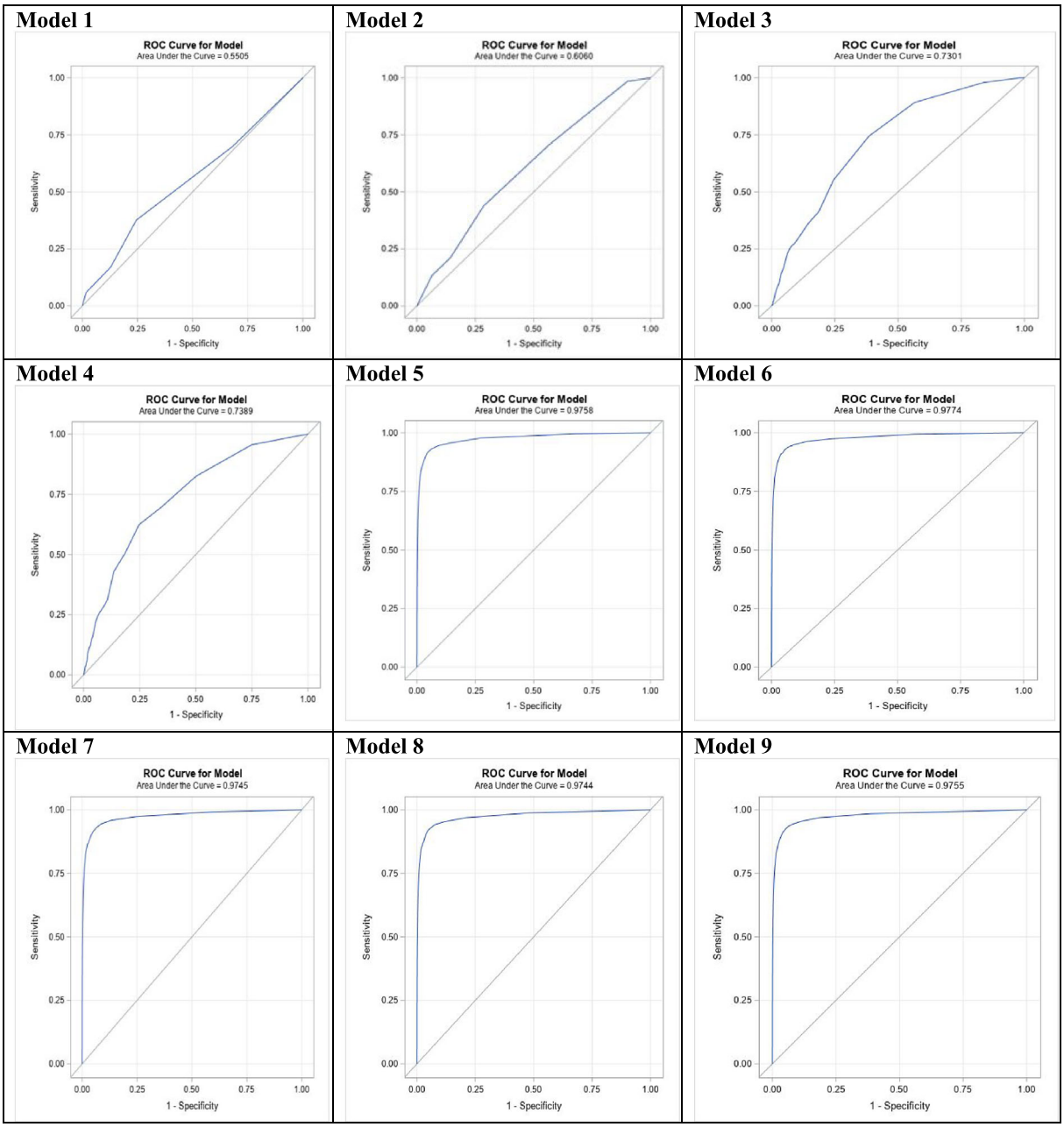

FIGURE 3 ROC curves of model specifications in Table 5. This figure presents the receiver operating characteristic (ROC) curves of the models in Table 5. The predictive accuracy of a model is measured by the area under the ROC curve. An area of 1 represents a perfect prediction; an area of 0.5 or below represents a worthless one.

with the in-sample estimated results reported in Panel A of Table 4 and confirm that failed banks in the first wave of bank failures are primarily smaller banks. In the presence of bank financial variables, the coefficient of *systemic funding cost* becomes statistically significant at the 5% level (model 6), while macroeconomic variables, either at national or state level, have no explanatory power (models 7 and 9). In contrast to the counterparts in Table 5, the coefficient of *bank excess return* is insignificant in all model specifications, which is sensible as *bank excess return* may have captured the performance of larger banks instead of privately held small banks. The results of Table 6 mostly confirm the main findings from Table 5.

Overall, our results indicate that bank-specific factors are the major determinants of bank failure; banking-industry factors also significantly affect bank failure. Macroeconomic conditions, particularly the state-level

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economic conditions, can significantly affect overall bank failures. However, models using either bank market variables alone or macroeconomic variables alone do not predict individual bank failures accurately as indicated by the AUROCs, and the incremental predictive power of macroeconomic variables is negligible in the presence of bank financial variables.

Our findings are consistent with existing empirical studies that suggest firm-specific characteristics are the major determinants of corporate default. Although taking the macroeconomic conditions into account may improve the prediction of corporate default or fragility, they also indicate that corporate default probabilities are driven by how firms are tied to the business cycle (Bonfim, 2009; Bruneau et al., 2012; Carling et al., 2007; Pesaran et al., 2006). Arena (2008) studies the relationship of bank failures and bank fundamentals during the 1990s Latin America and East Asia banking crises and finds that individual bank conditions explain the bank failures, while macroeconomic shocks that triggered the crises primarily destabilized the weak banks ex ante. Built on these existing empirical studies, in the next section, we examine the impact of macroeconomic conditions from a different perspective.

4.3 | Macroeconomic shocks and bank failure

In this section, we investigate how macroeconomic shocks affect bank failure through bank-level financial conditions. To address this issue, we interact macroeconomic variables with bank financial variables. The coefficients of the interaction terms will indicate how macroeconomic factors affect bank failure via the interacting bank financial variables.

An economic recession will increase the likelihood of default and therefore affect the performance of banks' loan portfolios. We construct a dummy variable, *Dgrowth*, that takes a value of one when the national GDP growth rate is negative, and zero otherwise. We then interact *Dgrowth* with *non-performing loan ratio (NPL)*. We expect that the coefficient of the interaction term, *Dgrowth*NPL*, will have a positive sign as an economic recession will worsen the quality of bank loans and thus increase the likelihood of bank failure.

As shown in the previous section, *systemic funding cost* (*sys_cost*) is a significant predictor of bank failure. However, it is not clear how systemic funding risk affects specific banks as this variable is the same for all banks. We interact this variable with the proxies for bank liquidity, *liquid asset ratio*, and *large CDs to assets*, respectively. Bank systemic funding cost reflects the market interest rates specific for banks. An increase of market interest rates may increase the refinance cost of banks, so we expect the coefficient of the interaction term between *sys_cost* and *large CDs to assets* will be positive. Since banks generally hold fixed-income securities as the major component of liquid assets, an increase of interest rates will have a negative impact on the market value of the investment securities. However, an increase of interest rates also increases the reinvestment income of banks. Therefore, the expected sign of the coefficient for the interaction term between *sys_cost* and *liquid asset ratio* is ambiguous.

Table 7 presents the estimated results. Models 1–3 report the results of adding *Dgrowth*, the interaction term between *Dgrowth* and *non-performing loan ratio* (*Dgrowth*NPL*), and both *Dgrowth* and *Dgrowth*NPL* into our baseline model (i.e., model 5 of Table 5), respectively. Models 1 and 2 of Table 7 show that the coefficients of both *Dgrowth* and *Dgrowth*NPL* are significantly positive at the 5% level. The results indicate that, given the same financial conditions of banks, the probability of bank failure is significantly higher during economic recession (model 1). Similarly, a national economic recession will exacerbate loan quality and significantly increase the likelihood of failure for those banks with higher non-performing-loan ratio (model 2). However, both variables lose their significance in predicting bank failure, as shown in model 3. A possible explanation is that banks' non-performing loans are generally higher during economic recession,[22] thus the marginal impact of *Dgrowth*NPL* given economic recession (*Dgrowth*), or the marginal impact of *Dgrowth* given the higher non-performing loans during recession period (i.e., *Dgrowth*NPL*), becomes insignificant.[23]

We next investigate the impact of *Dgrowth*NPL* on bank failures conditioned on macroeconomic variables including GDP growth (model 4), as well as both macroeconomic and banking-industry market variables (model 5), respectively. The coefficient of *Dgrowth*NPL* remains significant at 10% level, and the magnitude of the coefficients of both *NPL* and *Dgrowth*NPL* is smaller than their counterparts in model 2. The results indicate that a national economic recession will significantly increase the chance of failure in the following year for those banks with poor asset quality regardless of banking-industry factors and national economic conditions. However, we still need to bear in mind that its marginal impact conditioned on economic recession is insignificant as the probability of bank failure is significantly higher already during economic recession, as shown in models 1 and 3.

We add *systemic funding cost* (sys_cost), the interaction terms between *systemic funding cost* and bank liquidity variables, *sys_cost*Liquid asset ratio*, and *sys_cost*Large CDs to assets*, into the baseline model. As

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TABLE 7 Macroeconomic shocks and bank failure, 1977–2019.

Variables	1	2	3	4	5	6	7	8
Intercept	0.09	0.45	0.42	0.85	0.49	−0.55	0.23	0.18
Capital ratio	−53.98***	−54.57***	−54.65***	−55.17***	−56.28***	−53.42***	−56.18***	−56.19***
Earnings ratio	−10.28**	−9.87**	−9.74**	−9.34*	−8.74*	−12.18**	−8.76*	−8.75*
Non-performing loan ratio (NPL)	19.30***	16.61***	17.26***	16.35***	16.38***	19.71***	16.55***	16.65***
Other real estate	10.80***	10.86***	10.84***	10.52**	11.46***	11.74***	11.55***	11.60***
Liquid asset ration	−3.73***	−3.66***	−3.66***	−3.56***	−3.62***	−2.63	−2.80	−2.82
Large CDs to assets	3.08***	3.10***	3.12***	3.00***	2.83***	2.86**	2.75*	2.77*
Expense ratio	1.78	2.08	2.01	2.17	1.89	−0.74	1.89	1.88
Size	−0.19*	−0.21**	−0.22**	−0.23**	−0.23**	−0.16	−0.23**	−0.22**
Dgrowth	0.75**		0.33					0.12
Dgrowth * NPL		13.20**	9.82	11.78*	11.59*		11.00*	10.42
Sys_cost * liquid asset ratio						−94.28	−59.67	−57.36
Sys_cost * large CDs to assets						1.75	3.62	3.19
Bank excess return					−1.11*		−1.05*	−1.07*
Systemic funding cost (sys_cost)					19.34*	50.03*	34.62	33.29
Growth rate				−2.15	2.78		2.29	3.50
Inflation rate				−5.35	−8.00		−7.81	−7.67
Unemployment rate				1.57	3.96		4.38	4.27
AUROC	0.974	0.976	0.974	0.975	0.976	0.976	0.977	0.976
No. of failed banks	2015							
No. of observations	445,485							

Note: This table presents the estimation results using the time-varying hazard model. The results are estimated by using 1-year lagged bank financial variables, market and macroeconomic variables from 1976 to 2018 to predict bank failures over the period of 1977–2019. *Dgrowth* is a dummy variable based on national growth rates. The chi-square values for the hazard model have been adjusted for the average number of bank years per bank. AUROC is the area under the ROC curve.

***Indicates a significance level at 1%;

**Indicates a significance level at 5%;

*Indicates a significance level at 10%.

shown in model 6 of Table 7, the coefficients of the interaction terms have the expected sign but are insignificant, while the *liquid asset ratio* loses its significance and the coefficient of *systemic funding cost* is significant at only 10% level. When *Dgrowth*NPL*, *bank excess return*, and other national economic variables are added to model 6, as shown in model 7, *systemic funding cost* becomes insignificant while the coefficients of *Dgrowth*NPL* and *bank excess return* remain significant at the 10% level. The results imply that the marginal effect of *systemic funding cost* on bank failure via bank liquidity is not independent, but to a large extent depends on banking-industry factors as well as macroeconomic conditions. We include all variables in model 8, and as expected, *Dgrowth*NPL* loses its significance in the presence of *Dgrowth*, and vice versa.

Notably, the coefficient of *bank excess return* is negatively significant at the 10% level in all models (5, 7, and 8), consistent with the results in Table 5. The results indicate that, unlike *systemic funding cost* whose predictive power is subsumed in the presence of macroeconomic variables, our proxy for banking-industry market performance has a significantly independent predictive power on bank failures regardless of macroeconomic and market conditions. This finding is novel to existing bank failure literature and has important policy implications since information on bank stock returns is publicly available.

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5 | CONCLUSIONS

In this paper, we use the time-varying hazard model proposed by Shumway (2001) to investigate the impact of market and macroeconomic factors on bank failure in the U.S. over the sample period 1977–2019. We find that both market and macroeconomic factors can significantly affect overall bank failures. However, models using either banking-industry market variables alone or macroeconomic variables alone do not accurately predict individual bank failures, and the incremental impact of macroeconomic variables is insignificant in the presence of bank-level financial variables. Our results indicate that bank-specific factors are the major determinants of bank failure.

We do find that banking-industry market variables are significant predictors of bank failure even in the presence of bank-level financial variables. We confirm the finding of Hong and Wu (2013) that bank *systemic funding cost* is a significant predictor of bank failure, but provide new evidence that its predictive power depends on macroeconomic conditions as its significant predictive power vanishes once we include macroeconomic variables into the regressions. Unlike *systemic funding cost*, *bank excess return*, our proxy for banking-industry market performance, has a significantly independent predictive power on bank failure regardless of macroeconomic and market conditions. This finding is novel to existing bank failure literature and has important policy implications since information on bank stock returns is publicly available.

We also investigate how macroeconomic shocks affect bank failure via bank-level financial conditions. We find that an economic recession would significantly increase the failure likelihood of banks with higher ratio of nonperforming loans. Increase of bank systemic funding costs not only significantly increases the failure likelihood of banks but also makes those banks heavily relying on purchased funds more susceptible to failure, although the effect is not significant.

Our findings have important implications for bank regulators and policymakers as well as those engaging in individual bank risk management. First, our findings suggest that while bank-level financial conditions are more essential in predicting bank failure, banking industry market variables are also significant predictors. Thus apart from maintaining the effectiveness of the CAMEL supervisory framework, regulators may consider to incorporate banking-industry market information into the supervisory system to monitor bank failure, particularly the failure of larger banks. Second, our results demonstrate that an early warning model based on historical wave of failures can be quite effective at identifying troubled banks in subsequent periods. We find that the coefficients estimated from the in-sample period 1977–2005 (1985–1992) covering the first wave of bank failures correctly identify more than 92% (89%) of failed banks among the top 5% worst predicted probabilities in the out-of-sample period 2006–2019 (2009–2011). This indicates that the Federal Reserve Board's SEER model, developed in the early 1990s and documented in Cole et al. (1995), undoubtedly provided timely early warning to identify troubled banks at the start of the global financial crisis in 2008.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are mostly publicly available with detailed data sources provided in Section 2 of the paper. Data from Datastream International are subject to subscription restrictions.

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ENDNOTES

[1] During the 2008–2009 global financial crisis triggered by the collapse of Lehman Brothers, a contagious chaos in the global financial markets led to a global economic recession.

[2] The Uniform Financial Rating System, informally known as the CAMEL ratings system, was introduced by U.S. regulators in the November 1979 to assess the health of individual banks. Following an onsite bank examination, bank examiners assign a score on a scale of one (best) to five (worst) for each of the five CAMEL components; they also assign a single summary measure, known as the "composite" rating. In 1996, CAMEL evolved into CAMELS, with an addition of the sixth component ("S") to summarize Sensitivity to market risk.

[3] For 1976–2010, these data, along with supporting documentation for their use, are publicly available for download from the website of the Federal Reserve Bank of Chicago at: https://www.chicagofed.org/banking/financial-institution-reports/commercial-bank-data.

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For periods after 2010, these data are publicly available for download from the website of the U.S. FFIEC at: https://cdr.ffiec.gov/public/PWS/DownloadBulkData.aspx.

[4] These data are publicly available for download from the website of the U.S. FDIC at: https://www.fdic.gov/bank/individual/failed/banklist.html.

[5] These data are publicly available for download from the website of the BEA at: https://www.bea.gov/data/gdp/gdp-state

[6] These data are publicly available for download from the website of the BLS at: https://www.bls.gov/lau/data.htm.

[7] The data dictionary is publicly available from the website of the Federal Reserve at: https://www.federalreserve.gov/apps/mdrm/data-dictionary

[8] For example, total investment securities (TIS) is one of the data items needed to construct bank liquidity variable, and RCFD0390 is only available during the period 1984–1993. We then carefully follow the definitions of the data dictionary to calculate TIS: TIS=RCFD0400+RCFD0600+RCFD0900+RCFD0380-RCFD1000 for the period 1976–1983, and it equals to the sum of RCFD1754 and RCFD1773 for the period 1994–2018.

[9] Tier 1 equity capital and risk-weighted assets were not collected by U.S. bank regulators prior to 1994.

[10] *Past-due loans* refer to loans delinquent for 90 days or more but still accruing interest. Delinquent loans typically go into this category prior to transitioning into the more serious category of Nonaccrual loans.

[11] *Other real estate owned* refers to properties acquired by the bank by foreclosing on Nonaccrual loans and seizing underlying collateral. Other real estate owned is a non-earning asset item on banks' balance sheet, increasing other real estate owned of a bank's balance sheet may indicate the institution's credit is deteriorating.

[12] Other studies that use bank size to approximate the sensitivity to market risk include Avkiran and Cai (2012), Kolari et al. (2002), and Wheelock and Wilson (2000). Cole and White (2012) and Torna and DeYoung (2013) include bank size in their failure models, but do not connect it with sensitivity to market risk.

[13] We thank an anonymous reviewer for this suggestion. Our results are qualitatively unchanged when we use a winsorized sample.

[14] Thousands of banks also exited from the sample as a result of bank mergers and acquisitions.

[15] During the 1980s, banks were not allowed to open branches across state lines, and in many states, a bank was only allowed to have one location. In these "unit-banking" states, which included Texas and Illinois, bank holding companies would charter separate bank subsidiaries in multiple counties within a state. In many cases, the FDIC closed most or all of the subsidiaries of a bank holding company. In 1989, for example, regulators closed 20 of the 25 bank subsidiaries of MCorp—an $18 billion Texas bank holding company. At year-end 1985, there were 14,280 banks in the U.S. This number had declined to 7,219 by the year-end 2007.

[16] Intuitively, one can think of this as examining X% of all banks based upon the predicted probabilities of failure and assessing the percentage of all failures in that X% of all banks.

[17] Using a time-varying Cox hazard model, Wheelock and Wilson (2000) also find that the earnings ratio is not significant in predicting 1985–1993 U.S. bank failures.

[18] We are grateful to an anonymous reviewer for this suggestion.

[19] For example, Cole and White (2012) report that their model correctly identifies 96% of failed banks among the worst 5% of predicted failure probabilities, but they only use 2009 as an in-sample period and the first three quarters of 2010 as an out-of-sample period. Cole and Wu (2017) report an out-of-sample predictive accuracy of 92% among the worst 5% of predicted probabilities by using an in-sample period of 1985–1989 and an out-of-sample period of 1990–1993.

[20] We are grateful to an anonymous reviewer for this suggestion.

[21] The results are not reported here but are available upon request.

[22] We are grateful to an anonymous reviewer for this suggestion.

[23] The coefficient of a variable in the hazard model indicates the marginal effect of the variable given the conditions of all other independent variables in the model.

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